

Auteur: Thijs Van Schel
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$$f(x, y) = x^3 - x^2y + xy^2 - y^3$$

$a(1, -1)$ en $h(2, 3)$.

$$\begin{aligned} \left\{ \begin{array}{l} \frac{\partial f}{\partial x}(x, y) = 3x^2 - 2xy + y^2 \\ \frac{\partial f}{\partial y}(x, y) = -x^2 + 2xy - 3y^2 \end{array} \right. &\Rightarrow \left\{ \begin{array}{l} \frac{\partial f}{\partial x}(1, -1) = 3(1)^2 - 2(1)(-1) + (-1)^2 = 6 \\ \frac{\partial f}{\partial y}(1, -1) = -(1)^2 + 2(1)(-1) - 3(-1)^2 = -6 \end{array} \right. \\ \Rightarrow Df(a, h) = \begin{pmatrix} 6 & -6 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} &= -6 \end{aligned}$$

References